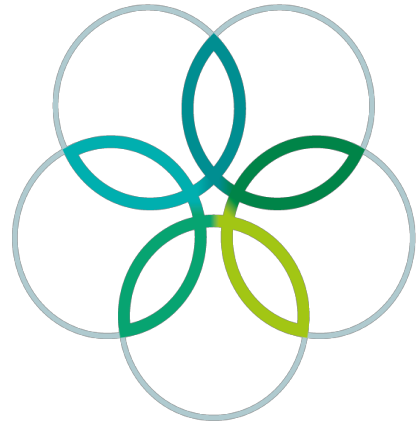
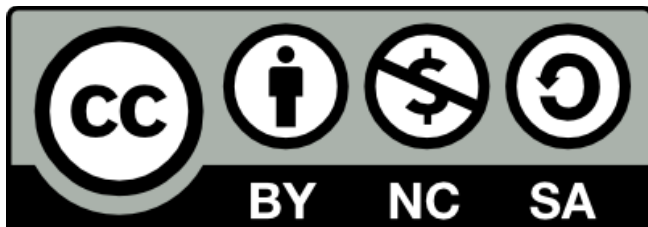


INTERNATIONAL
BIOLOGY
OLYMPIAD e. V.

IBO



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1. If you keep a C_3 plant and a C_4 plant of comparable size in an air-light glass container and provide them with adequate sun light, you would expect the C_4 plant to.

- a) grow better than the C_3 plant after one year in that condition;
- b) grow slower than the C_3 plant after one year in that condition;
- c) be able to survive longer than the C_3 plant, but ultimately die; +
- d) be able to survive not as long as the C_3 plant.

2. When compared to sun plants, shade plants usually:

- a) have leaves greener than those of sun plants because they need more chlorophyll;
- b) have thicker cuticles than sun plants because they need to preserve water;
- c) have thicker leaves because, this way, they can capture all the incoming light energy;
- d) have thinner leaves because they are already sufficient to capture low light energy. +

3. The process of photorespiration does not.

- a) occurs during the daytime;
- b) produce phosphoglycolate;
- c) consume oxygen;
- d) generate ATP. +

4. Which of the following genotype frequencies of AA, Aa and aa, respectively, satisfy the Hardy-Weinberg principle?

- a) 0.25, 0.50, 0.25; +
- b) 0.36, 0.55, 0.09;
- c) 0.64, 0.27, 0.09;
- d) 0.29, 0.42, 0.29.

5. When the base composition of DNA from bacterium *Mycobacterium tuberculosis* was determined, 18 percent of the bases were found to be adenine. What is the [G] + [C] content?

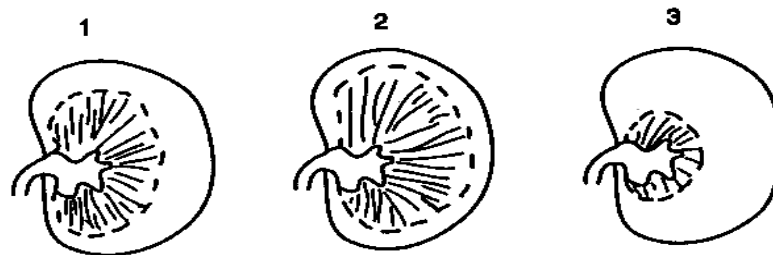
- a) 18%;
- b) 32%;
- c) 36%;
- d) 64%. +

6. Six tubes containing preparations from animal tissue were set up as shown in the table. After incubation, in which three tubes would carbon dioxide be produced?

N	content
1	glucose + homogenised cells;
2	glucose + mitochondria;
3	glucose + cytoplasm lacking organelles;
4	pyruvic acid + homogenised cells;
5	pyruvic acid + mitochondria;
6	pyruvic acid + cytoplasm lacking organelles.

- a) 1,2 and 3;
- b) 1,4 and 5; +
- c) 3,4 and 6;
- d) 3,5 and 6.

7. The diagrams show vertical sections of kidneys of coypu, brown rat and kangaroo rat, showing the relative size of cortex and medulla. Coypu are found in fresh water and are never short of water to drink. Brown rats are able to go some days without drinking. Kangaroo rats are able to live in deserts without drinking at all. Which kidney belongs to which animal?



- | | | | |
|----|--------------|--------------|---------------|
| | 1 | 2 | 3 |
| a) | brown rat | coypu | kangaroo rat; |
| b) | brown rat | kangaroo rat | coypu; |
| c) | kangaroo rat | brown rat | coypu; + |
| d) | kangaroo rat | coypu | brown rat. |

8. Rockmusicians, boiler-makers, and other workers who are subjected to loud, high-pitched noises over a period of years frequently become deaf to high tones because;

- a) the cells near the base of the organ of Corti become injured; +
- b) the tympanic membrane become injured;
- c) the membrane at the opening of the inner ear, called the oval window becomes injured;

d) the auditory nerve becomes injured.

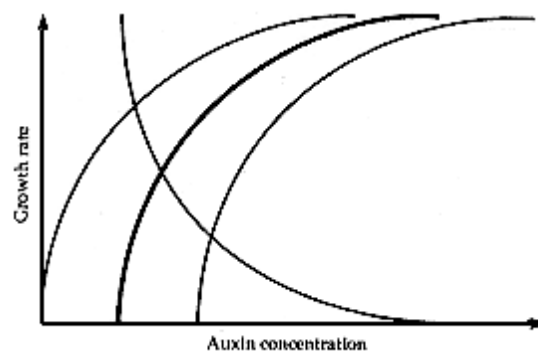
9. Characteristics of animals with open circulatory system.

- a) haemoglobin, haemocoel, lymph;
- b) haemocyanin, haemocoel, haemolymph;+
- c) haemoglobin, haemocoel is absent, haemolymph;
- d) haemocyanin, haemocoel is absent, lymph.

10. Corneal transplants are highly successful because.

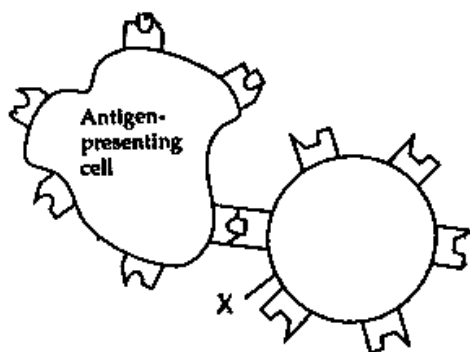
- a) the tissues from the patient and from potential donors are matched; +
- b) the cornea has almost no blood or lymphatic vessels associated with it;
- c) drugs and X-rays have been used to destroy T-lymphocytes;
- d) most patients do not have immune system.

11. The heavy line in the graph above illustrates the relationship between auxin concentration and cell growth in stem tissues. If the same range of concentrations were applied to lateral buds, what curve would probably be produced?



- a) I; +
- b) II;
- c) III;
- d) IV;
- e) either I or III.

12.



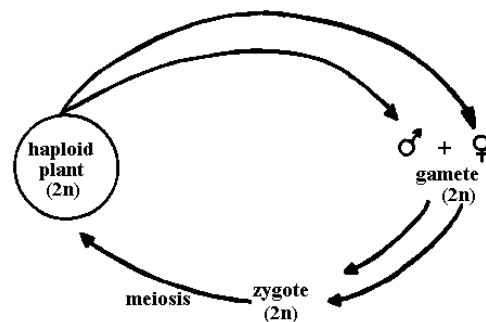
The cell represented by number "x" is

- a) B cell;
- b) T cell; +
- c) mast cell;
- d) macrophage;
- e) plasma cell.

13. When sunlight is on the chloroplast, pH is lowest in the

- a) stroma;
- b) cytosol;
- c) space enclosed by the thylakoid membranes; +
- d) space enclosed by the inner and outer membranes.

14.



What is the type of life cycle of green algae shown in the Fig. above?

- a) haplontic type; +
- b) diplontic type;
- c) isomorphic diplohaplontic type;
- d) heteromorphic diplohaplontic type.

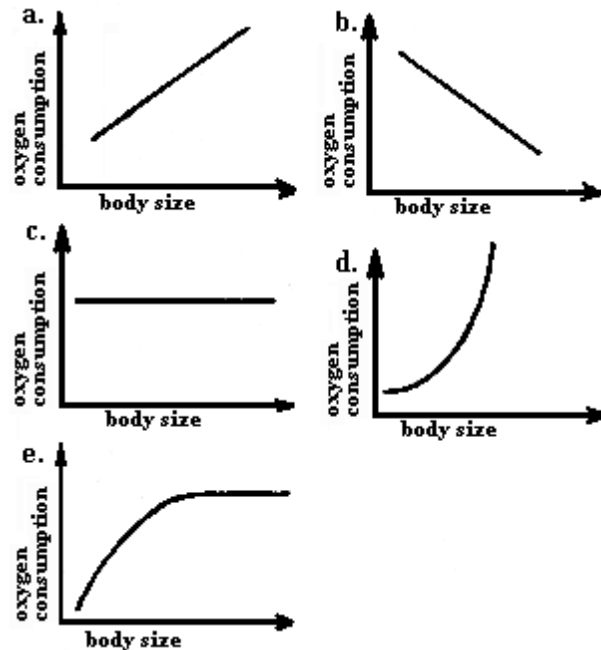
15. Red algae differ from the green algae and brown algae in having:

- a) no chlorophyll a;
- b) no differentiated cells;
- c) no phycocyanin within their cells;
- d) no flagellated stages in their life cycles. +

16. Ribosomes in the cytoplasm of eukaryotic cells are:

- a) the same size and composition as in bacteria;
- b) larger than in bacteria, but of similar composition; +
- c) smaller than in bacteria, and different in composition;
- d) the same size but completely different in composition from the ribosomes in bacteria.

17. The relationship between oxygen consumption and weight (body size) of sea anemones was studied. The rate of oxygen consumption depends on the surface area of the animal. Which of the following graphs best describes this relationship?



- a) A;
- b) B;
- c) C;
- d) D;
- e) E.

18. Most metabolism of a plant is carried out by the

- a) epidermis;
- b) collenchyma;
- c) sclerenchyma;
- d) parenchyma. +

19. Tissues that form long, tough strands, as in the leaf stalk of celery, are:

- a) epidermis;
- b) collenchyma;
- c) sclerenchyma; +
- d) parenchyma.

20. The lowest water potentials in the xylem are in the

- a) root hairs;

- b) vascular cylinder of roots;
- c) tracheids of the stem;
- d) leaves. +

21. Many fungi look like and live like plants, but they are put in a separate group because:

- 1. Fungi are prokaryotic;
- 2. Many fungi are parasitic;
- 3. Fungi do not have embryological development;
- 4. The cell wall of fungi do not contain cellulose;
- 5. sexual reproduction is lacking in many groups of fungi.

- a) 1,3,5;
- b) 2,4;
- c) 3,4,5;+
- d) 4,5.

22. Which of the following features are shared by the Eubacteria?

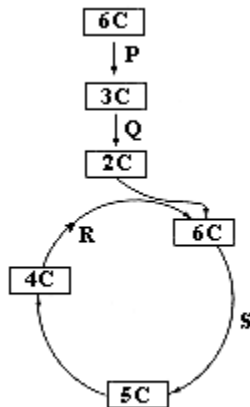
- 1. prokaryotic cell;
- 2. rigid cell wall with peptidoglycan;
- 3. eukaryotic cell;
- 4. multiple by binary fission;
- 5. rigid cell wall with cellulose;

- a) 1, 2, 4; +
- b) 2, 3, 4;
- c) 3, 4, 5;
- d) 1, 2;
- e) 3, 5.

23. One of the most widely used natural insecticides are the toxins produced by

- a) *Bacillus thuringiensis*; +
- b) *Salmonella* sp.;
- c) *Bacillus stereothermophilis*;
- d) *Bacillus cereus*;
- e) *Escherichia coli*.

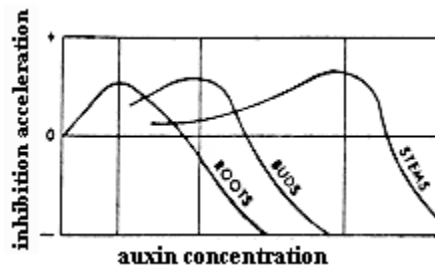
**24. The diagram represents the respiratory pathway
Which stages involve the action of both a decarboxylase and a dehydrogenase?**



- a) P and Q;
- b) Q and R;
- c) Q and S; +
- d) R and S.

25. For the next ten items refer to the data plotted on the following graph.

Use statements, which are supported by the data of this graph.



1. Stems do not respond to auxins.
2. Plant roots do not respond to auxins.
3. Roots react differently to auxins than stems do.
4. Roots respond to different auxins not so as stems do.
5. Roots excrete auxins and thus protect themselves.
6. High concentration of auxins increase root growth.
7. Stem growth is always increased by the addition of auxins.
8. The growth of a root is promoted by amounts of auxins smaller than those for stem growth.
9. Root growth is inhibited by stem growth.
10. Root growth is accelerated by stem growth.

- a) 3, 8; +
- b) 1, 2, 6, 7;
- c) 4, 5, 9, 10;
- d) 3, 6, 10;
- e) 1, 8.

26. The following are statements about the female reproductive system. Respond by true.

1. Both oestrogen and progesterone are necessary for ovulation to take place;
2. Oestrogen tends to inhibit the production of FSH by the anterior pituitary gland;
3. Fertilization of the ovum by the spermatozoon normally takes place in the uterus;
4. Progesterone production is largely under the control of Lh;
5. Throughout the part of the mensual cycle that follows ovulation, there is a slight rise in body temperature.

- a) 1, 2, 5;
- b) 1, 3, 5;
- c) 2, 3, 4;
- d) 2, 4, 5;+
- e) 5.

27. A man of blood group A has 2 children. Plasma from the blood of one

of them agglutinates his red cells while that from the other does not.

Choose correct statements.

1. Father must be heterozygous group (A).
2. Children must have different mothers.
3. "Agglutinating" child could be group O.
4. Mother of "agglutinating" child must be group O.
5. "Non-agglutinating" child could be group AB.

- a) 1, 2, 5;
- b) 1, 3, 5;+
- c) 2, 3, 4;
- d) 2, 4, 5;
- e) 5.

28. Ultrafiltration is responsible for the fluid movement that takes place in the following processes:

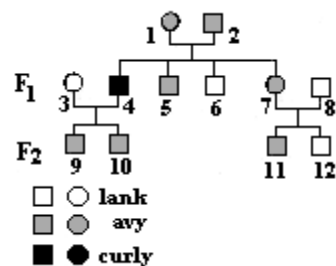
- a) concentration of bile;
- b) salivation;
- c) glomerular filtration; +
- d) sweating;

29. If (Rh⁺) blood is transfused into an (Rh⁻) woman who has not previously been transfused, then:

(Choose correct statements.).

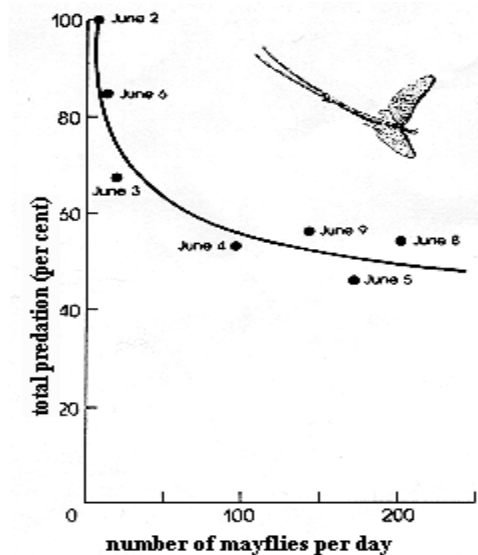
- a) anti-Rh antibodies will be produced in the organism of women;+
- b) the blood is incompatible so red cell agglutination and death may follow;
- c) there is no immediate or long term effect as 70% of the Rh population are heterozygous;
- d) provided anti-D antibody is given before the next pregnancy no harm will be done.

30. The next four items are based on the family tree below. The studied character concerns the hair. Choose incorrect statements.



1. The character wavy hair is dominant
 2. The number one is homozygote
 3. One of the parents of the N2 had probably the same genotype as him
 4. If number 3 and number 4 could have another child he should have lank hair.
- a) 1, 3;
 - b) 2, 4;
 - c) 1, 2, 4;+
 - d) 2, 3, 4;
 - e) 1, 2, 3, 4.

31.



Mayflies are tasty aquatic insects that are especially vulnerable to predator during their transition to adult hood, when they are emerging from the water to mate and lay their eggs. This graph illustrates the effect of the number of mayflies emerging per day on the total predation on them.

Choose correct statement.

- a) the more mayflies emerging on a June evening, the more likely any one mayfly is to be taken by a predator. -
- b) the more mayflies emerging on a June evening, the less likely any one mayfly is to be taken by a predator. +
- c) the number of emerging mayflies diminishes steadily from the beginning to the end of June. -

32. Oxygen content reduction makes the glycolyse (glycogenesis) intensity increased due to:

- a) increase of ADF concentration in cell; +
- b) increase of NAD^+ concentration in cell;
- c) increase of ATP concentration in cell;
- d) increase of concentration of peroxides and free radicals.

33. Nucleosomes prevent:

- a) the replication and all the stages of DNA-transcription;
- b) the beginning of DNA-transcription, but don't prevent it's continuation and replication; +
- c) the replication and the beginning of transcription, but don't prevent it's continuation;
- d) the continuation of the transcription, but don't prevent it's beginning and replication.

34. The main organ which produces glucose from lactic acid is:

- a) kidneys;
- b) liver; +
- c) spleen;
- d) intestines epithelium.

35. Ribosomes consist of:

- a) RNA and proteins; +
- b) RNA, proteins and lipids;
- c) lipids and proteins;
- d) RNA, proteins, lipids and carbohydrates.

36. The movement of flagellum of eukaryotic cells realised by means of hydrolusis of ATP by:

- a) actin;
- b) myosin;
- c) dynein; +
- d) kinesin;

37. Flagellum of procaryots moves by means of the energy of:

- a) ATP;
- b) electrohimical potential;
- c) phosphorylation;
- d) phosphoendpiruvate;

38. Genetic material transmission from one bacterium to another one with the help of virus is called:

- a) transposition;
- b) transformation;
- c) transversion;
- d) transduction. +

39. Inner lysosomes medium is:

- a) more acid than that of cytoplasma;+
- b) more alkaline than that of cytoplasma;
- c) has the same pH that is in cytoplasma;
- d) sometimes more acid and sometimes more alkaline.

40. HIV agglucts:

- a) T- lymphocytes; +
- b) B-lymphocytes;
- c) monocytes;
- d) neutrophilus;
- e) basophilus.

41. A heron standing in a cold water for a long time doesn't get its legs overchilled because of:

- a) countercirculation in limbs ; +
- b) even thin fat layer under limbs'skin;
- c) branched blood stream in limbs:
- d) intensive metabolism in limbs.

42. Decrease of excitability of nerve cells is:

- a) membrane potential decrease after absolute magnitude;
- b) change of rest potential of a cell towards negative; +
- c) membrane potential change from negative figures towards positive ones;
- d) changing of membrane potential sign from positive to negative.

43. It is possible to incubate in agar the culture of agents of:

- a) diabetes;
- b) influenza;
- c) malaria;
- d) dysentery. +

44. Which structure of the mammals is a modified "parietal eye"?

- a) hypophysis;
- b) cerebellum;
- c) epiphysis; +
- d) diencephalon.

45. The following creatures are natural carriers of the agent plague of human being in the nature:

- a) wolves, foxes;
- b) birds;
- c) rodents; +
- d) a human being.

46. Which group of modern reptiles is the most ancient:

- a) lizards (Sauria);
- b) snakes (ophidia);

- c) turtles (chelonina); +
- d) crocodiles (crocodilia);

47. Ectoderm, endoderm and mesoderm develop into tissues and organs. Which of the following combinations is correct?

	Ectoderm	mesoderm	endoderm
a)	nervous system	blood	lungs; +
b)	nervous system	large intestine	lungs;
c)	skin	bones	kidneys;
d)	epidermis	liver	heart.

48. The corn culm diameter is determined by the activity of the following type of meristem:

- a) primary; +
- b) secondary;
- c) primary and secondary;
- d) first primary then secondary.

49. Secondary thickening of a stem is typical for:

- a) mosses, gymnosperms, angiosperms;
- b) monocotyledonous, angiosperms, gymnosperms;
- c) monocotyledonous and dicotyledonous angiosperms;
- d) gymnosperms and dicotyledonous angiosperms. +

50. All gelminthes have such distinctive features as:

- a) absence of digestive system;
- b) no sense organs;
- c) hermaphroditism;
- d) high reproduction potential.

51. Mammals arterial blood flows through veins and venous blood through arteries:

- a) in the systemic circulation;
- b) in the pulmonary circulation; +
- c) in liver's portal system;
- d) during extrasystolic circulation when the blood is being pumped from heart's ventricle into auricles.

52. Why do man's eyes constantly move while looking at an object?

- a) to keep the object in the field of vision;
- b) to provide the direction of light beams to the yellow spot of retina;
- c) to keep the light on the receptors of an eye for a short time;
- d) to provide focus of image on the retina;+
- e) to deadaption of optic neuron.

53. Which of the following fossil plants could be ancestors of the contemporary highe plants?

- a) Psilophyta;+
- b) Rhuniophyta;
- c) Phaeophita;
- d) Rhodophyta;
- e) Cylorophyta.

54. Which of the following higher plants was brown coal formed from?

- a) jointweeds;
- b) ferns;
- c) lycopodiums;
- d) gymnosperms;
- e) rborescentflowering plants.

55.Which of following functions couldn't be considered as the functiopn of human digestion system?

- a) physical processing of food;
- b) hydrolysis of food components to monomereses;
- c) elimination with species specificity of food components;
- d) to free the energy during the oxydation of food components;+
- e) antybacterium processing of food.

56. Primary lysosomes are produced in:

- a) cytoplasm;
- b) nucleus;
- c) canalicular apparatus;+
- d) cell center.

57. Proteins that have been produced in the rough endoplasmatic reticulum are transported to:

- a) lisosoms;
- b) hyaloplasm;
- c) mitohodrium;

- d) plastids;
- e) canalicula apparatus.+

58. Gibberellins (substances of plants growing) are:

- a) phospholipids;
- b) three acyl glycerol;
- c) wax;
- d) steroids;
- e) terpenes.+

59 Gens for ribosomal RNA in chromosomes are situated in the area of:

- a) telomere;
- b) primary stretching;
- c) kinetochore;
- d) satellite;
- e) secondary stretching.+

60 What determines the connection between ribosomes and reticulum?

- a) carbohydrates;
- b) N-terminal protein sequence;+
- c) C-terminal protein sequence;
- d) lipids;
- e) ions.

61. What is the main factor to limit the increasing of general biomass on the definite territory?

- a) deficiency of O_2 and CO_2 ;
- b) deficiency of H_2O ;
- c) temperature;
- d) flow of sun energy;+
- e) biotyc relations.

62. Spatial structure of actin-myosin system is put into the basis of the muscular tissues classification. According to this way of approach all muscular tissues of multicellular are divided into three types:

- a) non-striated, cardial, cross-striated
- b) non-striated, skew-striated, cardial;
- c) non-striated, skew-striated, cross-striated;+
- d) epithelium-muscular, symplastic, cardial;
- e) postural, phasic, paramyosinic.

63. Heart systole force is known to quickly change when the concentration Ca^{2+} in the extracellular liquid is changed. Removal Ca^{2+} from perfusive solution leads to stopping of heart activity. It's connected with:

- a) stopping of generation of action potentials in pacemakers;
- b) stopping of generation of action potentials in myocard;
- c) Action potentials are generated in pacemakers but excitation transmission is disturbed in the heart;
- d) action potentials are generated in the heart but electrochemical conjugation is corrupted;+
- e) action potentials are generated in the heart but excitation transmission is disturbed via.

64. There are some ways to define the increasing of speed of excitation transmission via axon. The main of them are:

- a) diameter of axon;
- b) distance between Ranvier's interception and amplitude of action potential;
- c) magnitude of rest potential and size of;
- d) a.&b;+
- e) a.&c.

65. In nature there are only several main mechanisms of moving which are used by different groups of animals in different ways. This is:

- a) walking, flying, swimming;
- b) ciliary, flagellate, undulating;
- c) muscular, ciliary, amoeboid;+
- d) muscular, reactive, amoeboid;
- e) muscular, undulating, reactive.

66. The high limit of force (in relative magnitudes) is approximately the same for locomory muscles of different animals. It can be explained in following way:

- a) the force of muscle doesn't depend on area of cross-section;
- b) the force of the muscle doesn't depend on it belonging to phasic or postural groups of muscles;
- c) the force of muscles contraction depends on quantity of myofibrillas that take part in contraction;+
- d) the force of muscles contraction depends on reserve of ATP in muscles cell;
- e) the force of muscles contraction is limited by entrance of O_2 .

67. The heart of the homoiotherms has the mechanical structures (endoskeleton) which formed by fibers of collagen and elastic. It's function is:

- a) to provide the length of rest of the cardiac muscle and to prevent its oversprain;
- b) to increase efficiency of the cardiac muscle and to provide it's quick returning to the primary condition;
- c) to provide active affluence of blood to the heart;
- d) all answers are correct;+
- e) there are no correct answers.

68. Which animals have the system of giant axons?

- a) Coelenterata, Plathelminthes, Nemathelminthes;
- b) Plantheminthes, Nemathelminthes, Annelida;
- c) Nemathelminthes, Annelida, Artropoda;
- d) Annelida, Artoropoda, Mollusca;+
- e) Artropoda, Mollusca, Chordata.

69. Which reactions are made with the help of system of giant axons?

- a) quick non differential reactions;+
- b) quick differential reactions;
- c) slow non differential reactions;
- d) slow differential reactions;
- e) a.&b.

70. In human stomach splitting of the great bulk of carbohydrates is realised with the help of:

- a) amylase pH 5,8-7,8;+
- b) amylase pH 1,5-1,8;
- c) lactase pH 5,8-7,8;
- d) lactase ph 1,5-1,8;
- e) theres no correct answer.

71. How is increasing of temperature up to an optimum level (for graving out of pollen to embryo bag of pistic inside flowers of Alpine plants) can be reached in conditions of rather low temperature of air?

1. Flowers of plants are dark-coloured and are heated up by the sun well;
2. Petals of these flowers are closed a large part of their existence and consequently keep warm better.
3. Flowers of these plants have a high level of breath, which is accompanied by allocation of energy of metabolism as ATF; +
4. Petals of the majority of these flowers, similiary to parabolic mirrors, concentrate solar beams, in the centre of the flower where generative organs (pistic).+

- a) 1;
- b) 1, 2;
- c) 2, 3;
- d) 3, 4;
- e) 1, 2, 3, 4.

72. Explain, why do many granivorous birds (sparrow, goldinch) feed nestlings not by a grain, but by insects.

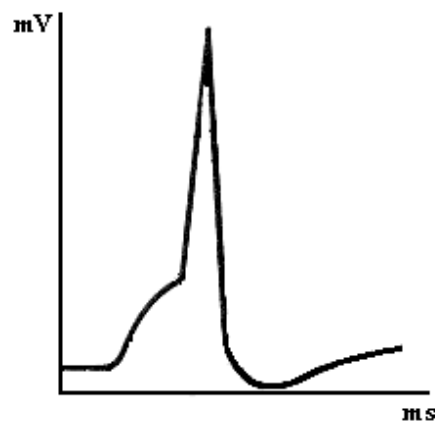
- 1. grain and seeds don't ripen to time rearing of nestlings.
- 2. nestlings digest a grain poorly;
- 3. animal food is more full value for nestlings.

- a) 1;
- b) 2;
- c) 3;
- d) 1, 2;
- e) 2, 3.+

73. Competitive inhibitors of ferments have:

- a) similar structure with a molecule of ferment;
- b) similar structure with a molecule of the substratum;+
- c) similar structure with the gene, which codes the ferment;
- d) ability to connect a molecule of the ferment and the substratum;
- e) ability to inhibit ferment.

74. The graph below illustrates action potential of the neuron. Using the code denote which parts of it are marked with letters.



The codes of the answers:

- 1. repolarization;
- 2. rest potential;
- 3. depolarization;
- 4. after hyperpolarization;
- 5. paradoxical phase;

6. levelling phase;
7. inhibitory phase;
8. parabiosis;

- | | A | B | C | D |
|----|----|----|----|-----|
| a) | 7; | 4; | 3; | 1; |
| b) | 1; | 3; | 2; | 5; |
| c) | 2; | 3; | 1; | 4;+ |
| d) | 5; | 6; | 3; | 7; |

75. Choose the features that confirm the hypothesis of endosymbiosis in respect of beginning of chloroplastids.

1. single - membrane,organelles;
2. double - membrane organelles;
3. they have the genetic make up of their own that includes circular DNA;
4. they have no the genetic apparatus of its own;
5. they have ribosomes that differ from ribosomes of cytoplasme in structure;
6. they have ribosomes that do not differ from ribosomes of cytoplasm;
7. they have a lot of common features with bacterium;
8. they have a lot of common features with blue - green algae.

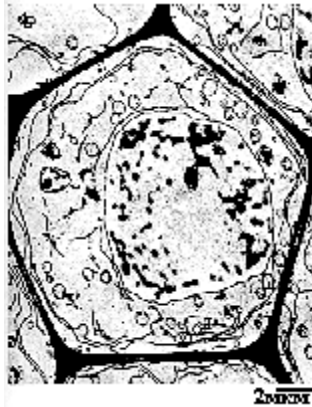
- a) 1, 4, 5, 7;
- b) 2, 4, 6, 7;
- c) 1, 3, 6, 8;
- d) 2, 3, 5, 8;+
- e) there are no correct answers.

76. Which of the following characteristics belong to electrical synapses?

1. They provide quick reactions;
2. They provide slow reactions;
3. They provide synchronous responses;they provide nonsynchronous responses;
4. They provide the coordination of cells that carry out the same functions;
5. They provide the coordination of cells that carry out different functions;
6. Vertebrates have them;
7. Chordates have them.

- a) 1, 2, 6, 8;
- b) 1, 3, 5, 7, 8;
- c) 2, 4, 6, 7, 8;
- d) 1, 3, 5, 7;
- e) 2, 4, 6, 8.

**77. Diagram I shows an electron microscopic picture. What is depicted?
Please mark the correct answers!**



1. A cell;
2. A tissue;
3. An organ;
4. A plant organism;
5. An animal organism;
6. A procaryotic cell;
7. An eucaryotic cell;
8. A fatty tissue;
9. A supportive tissue;
10. A cross-section of a root;
11. A cross section of a plant axis;

- a) 3, 4, 11;
- b) 2, 9, 10;
- c) 1, 4, 6;
- d) 1, 4, 7;+
- e) 3, 5, 8.

78. The plasmatic membrane of all cellss consists of:

- a) lipids, proteins and carbohydrates;+
- b) only lipids;
- c) only proteins;
- d) lipids and carbohydrates;
- e) lipids and proteins

79. As early in the XI and XIIth century knowledgeable healers recommended applying soil gathered from beneath black alders *Alnus glutinosa* to inflammed skin. Please state the reason for their success.

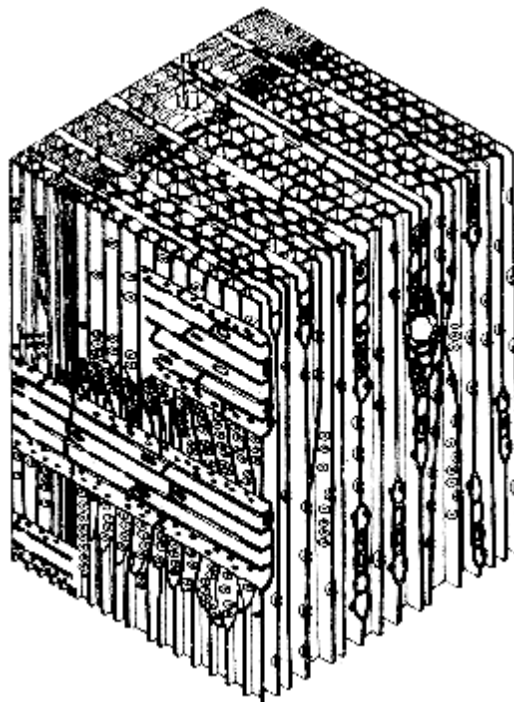
- a) black alders form a symbiotic relationships with a certain kind of fungus. Together they form small root bulbs to assimilate nitrogen which are called mykorrhiza. This mykorrhiza is also inhabited by streptomycetes which live near black alder roots. They exude streptomycin. This is responsible for the healig effect. +

- b) different kind of penicillium flourish in the soil beneath black alders.
Penicillin, which is dissolved due to the ground humidity, is responsible for the healing effect.
- c) the soil around black alders is relatively sterile due to the substances contained in the roots. If the soil is applied to a wound, pathogens are deterred.

80. Microbiologic culture media are produced in a pressure cooker at a temperature $+121^{\circ}\text{C}$. Which of the following organisms survive the procedure of simply boiling the culture media at $+100^{\circ}\text{C}$?

- a) *Escherichia coli*;
- b) *Penicillium notatum*;
- c) *Bacillus subtilis*; +
- d) *Salmonella typhimurium*;
- e) *Convallaria mayalis*.

81. Regard the following illustration and decide what kind of structure is depicted.



- a) The depicted structure is the wood of an angiosperm plant, since the strands of living tissue are made up of several layers;
- b) The depicted structure is the wood of a gymnosperm plant;+
- c) The depicted structure is the wood of an angiosperm plant, since it possesses vessel elements;
- d) The depicted structure is the wood of an gymnosperm, since the strands of living tissue are single layered.

82. Accessory pigments of photosynthesis transfer the absorbed energy to photosynthetically active pigments, hereby losing more or less considerable amount of energy. Which of the below mentioned substances do not qualify as accessory pigments?

- a) chlorophyll a;+
- b) chlorophyll b;
- c) carotenes;
- d) fucoxanthines of phaeophyceae;
- e) phycobiliproteids.

83. Many insects go into diapause at certain times. Which of the following statements is incorrect?

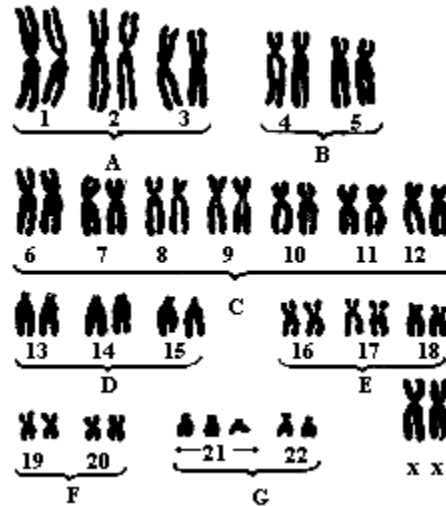
- a) the rate of metabolism is decreased to a minimum during diapause;
- b) the growth rate is stagnant during diapause.
- c) factors such as a lack of oxygen, vitamin deficiency, lack of nutrition etc. could effect a stagnation of growth during diapause.
- d) diapause is only caused by a shift in temperature and the duration daylight hours.+
- e) even the eggs of insects can go through diapause phases.

84. In situation that conflict between attack flight animals have been noted to behave in a most peculiar fashion eg. the behaviour of fighting cocks interrupting their fight to peck at the ground for food. What is this kind of behavior called and what this behavior is based on?

- I. Feeding
- II. Ritualisation
- III. Displacement Activity+
- IV. Aggressive Behaviour
- V. lack of food or hunger
- VI. lateral inhibition of two opposing behavior tendencies
- VII. irritation of a nerve
- VIII. displacement of an initial stimulus from one nerve to another

- a) I and V;
- b) II and VII;
- c) III and VII;
- d) III and VI;+

85. The following diagram depicts the human caryogram with a hereditary disease. What is the syndrome called? Which of the below mentioned conditions is caused by the which deficiency?



- I. Edwards syndrome
- II. Down syndrome
- III. Turner syndrome
- IV. Slanted eyes, skeletal deformity (flattened back of the head) , severe mental retardation, male or female.
- V. Feminine constitution, rudimentary and functionless sexual organs, infantilism.
- VI. Deformed fingers, heart conditions, high voices, lessened beard growth.

- a) I and V;
- b) II and IV;+
- c) II and IV;
- d) II and V.

86. The following statements are possible, general connections between the speed of evolution and sexual or asexual reproduction.

- I. The rate of evolution will be higher in a smaller population than it would be in a larger one.
- II. The rate of evolution will behave oppositely proportional to the generative duration.
- III. The rate of evolution will increase in proportion to the variety of the different ways of life. It will decrease in proportion to the density of competition which newcomers find in an certain area.

- a) only III is right;
- b) only I and II are right;
- c) all statements are correct;+
- d) none of the statements are correct;

87. The rate of hereditariness H can expressed in the following mathematical manner. Which statement is correct?

$$\text{a) } H = \frac{V_G}{V_P} = \frac{V_G}{V_G + V_E}$$

$$\text{b) } H = \frac{V_P}{V_G} + \frac{V_P}{V_E}$$

$$\text{c) } H = \frac{V_G}{V_G + V_P}$$

$$\text{d) } H = \frac{V_G}{V_P + V_E}$$

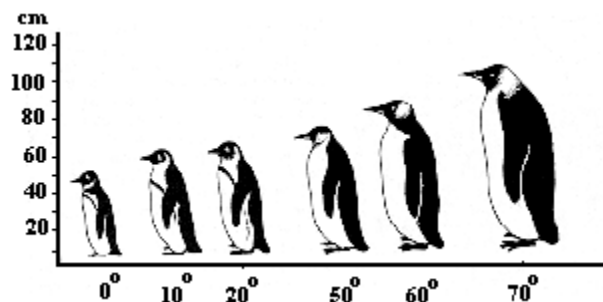
V_G genetic variability;

V_P phenotypic variability;

V_E environmentally caused (modified) variability.

- a) A;+
- b) B;
- c) C;
- d) D.

88. The connections between organism and their environment are diverse and numerous. The following illustration depicts a comparison of height between six different kinds of penguins. Please comment on the picture. Which factor (I-IV) determines the differences in the penguins' size? Which rule (V-VIII) describes these circumstances?

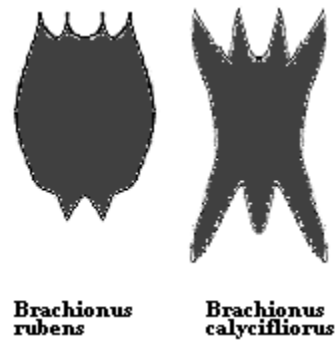


- I. The nutritional supply;
- II. The temperature;
- III. The competition to other species;
- IV. The different levels of oxygen in their native waters.
- V. The pessimum law;
- VI. The Allen rule;
- VII. The Bergmann rule;
- VIII. The ecological potency

- a) I and III;
- b) I, II and VIII;

- c) II and VII;+
- d) IV and VIII.

89. *Brachionus calyciflorus* is a frequent inhabitant of eutroph ponds and lakes. Different kinds are known, especially one with a pair of thorns on the hind or caudal end of the shell. This appearance, which is quite frequent with plancton organisms is known as cyclomorphosis:



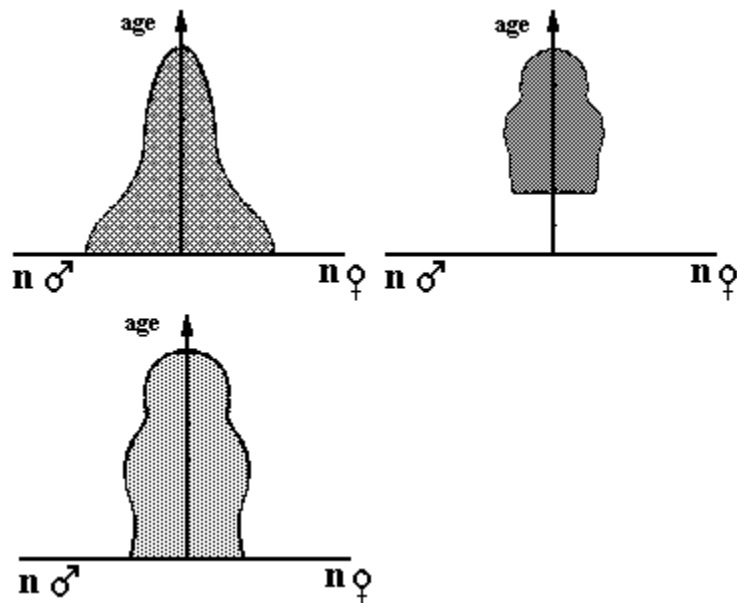
Brachionus rubens does not possess thorns on its hind end as a means of defense.

A different kind of single cell organisms, *Asplanchna brightwellii*, feeds amongst others on *Brachionus calyciflorus* and *Brachionus rubens* by attacking them from the rear in order to devour the prey. How would the population density develop if you mixed a culture of *Asplanchna brightwellii* with *Brachionus calyciflorus*? How would the population density develop if you mixed a culture of *Asplanchna brightwellii* with *Brachionus rubens*?

- I. The predator would survive;
- II. Predator and prey would survive;
- III. The prey would survive.
- IV. The predator would survive;
- V. The prey would survive;
- VI. The population would develop according the Volterra laws.

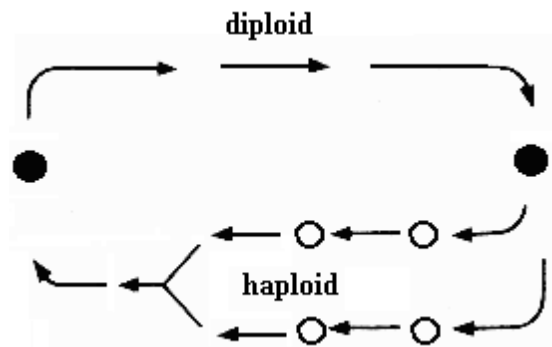
- a) I and IV;
- b) II and VI;
- c) III and VI;+
- d) III and V.

90. The diagrams A to C show population pyramids of a bee populations in different seasons (spring, summer and winter). Please match the spring with the corresponding illustration.



- a) A;+
- b) B;
- c) C.

91. Reproduction is often connected with a shift from haploid to diploid or the other way round. Please regard the diagram and decide with group of organisms show the below mentioned type of shift.



- a) eg. for humans;
- b) eg. for angiosperm plants;
- c) eg. for gymnosperm plants;
- d) eg. for and ferms.+

92. The following dichotomous key will be useful in determining the differences between cereals. At which point in this key would you place wheat (*Triticum*)?

1. Cereal ear without stem with a very short stem - organism A
- 1*. Cereal ears have long stems.
2. On every nodium of the cereal head's axis there is more than on ear - organism B.
- 2*. On every nodium of the cereal head's axis there is only one ear.
3. Ears contain three or more blossoms - organism C.

3*. Ears contain two blossoms.

- a) A;
- b) B;
- c) C.+

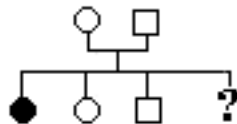
93. Age of some trees can be determined due to the presence of the "tree-rings" (annual growth rings) , which represent the yearly growth of:

- a) primary phloem and xylem;
- b) secondary phloem and xylem;
- c) secondary phloem only;
- d) secondary xylem only;+
- e) cork.

94. The eye of a squid and the eye of a fish are an example of:

- a) secondary adjustment to life in a water habitat;
- b) vestigial organs;
- c) homologous organs;
- d) analogous organs;+
- e) two of the above answers are correct.

95.The pedigree represents a women suffering from a certain rare disease caused by a recessive mutation in one gene. What is the probability that fourth child in the family (marked by a"?") is also ill? The mutation in question behaves according to classical Mendelian genetics.



- a) zero;
- b) 0.25;
- c) 0.5;
- d) 0.125;
- e) 0.25 if the gene is autosomal, 0.125 if it is on a sex chromosome.

96. On illumination green plants release oxygen gas. Oxygen atoms derived from:

- a) reduction of CO₂;
- b) decomposition of water;
- c) decomposition of sugars;
- d) photorespiration;
- e) two of the above possibilities may be correct under different circumstances.

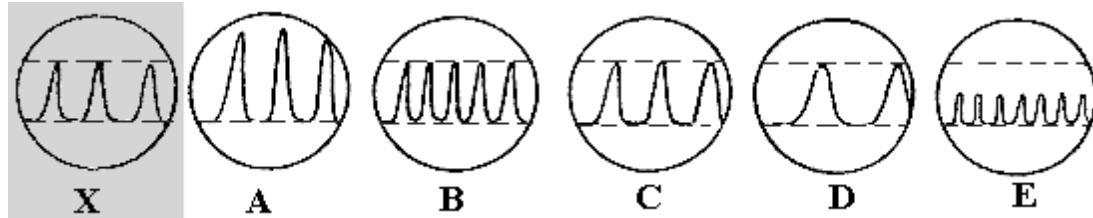
97. In root nodules, the nitrogen-fixing bacteria obtain from the host plant:

- a) nitrogen dissolved in vascular sap;
- b) oxygen dissolved in vascular sap;
- c) nitrates
- d) sugars
- e) neither of the above;

98. Certain physiological process occurs in a plant only after it has been illuminated with full-spectrum white light or its red component; other monochromatic cannot produce this effect. Based on this, one may expect that process is regulated via:

- a) chlorophyll;
- b) a flavonoid pigment;
- c) phytochrome;
- d) auxins or giberellins;
- e) the information given is not sufficient to choose from among above answers.

99. A medium-strong stimulation of a tactile receptor (touch receptor) causes in its nerve fibre an action potential, whose oscillograph representation is shown in fig. "X". How will the action potential look like after a stronger stimulus?



- a) A;
- b) B;
- c) C;
- d) D;
- e) E.

100. After marathonrun oxygen debt has formed in muscles of the runner;

a rest period after the run, which of the following will take place in the runners organism with particular intensity?

- a) conversion of pyruvate into lactate;
- b) conversion of lactate into pyruvate;
- c) accumulation of NADH;
- d) glycolysis;
- e) two of the above;

101. Is it possible that individual cells of a certain species of bacteria harbor different numbers of a plasmid molecule?

- a) no, only one plasmid molecule may be present per cell;
- b) yes, but only if are different kinds of plasmids;
- c) yes, but only if the bacteria are cultivated differing conditions;
- d) yes, because replication of most plasmids is not very strictly controlled;
- e) yes, but only for different strains of the bacterium.

102. Which of the features characteristic for (most) mitochondria and chloroplasts is NOT indicative of the organelles' endosymbiotic origin:

- a) possession of own, usually circular DNA, and own ribosomes;
- b) structure and composition of the organelles' ribosomes;
- c) the way these organelles multiply;
- d) presence of numerous introns in the organelles' genes;
- e) all of the above features reflect the similarity to prokaryotes, therefore support the endosymbiotic theory of the organelles' origin.

103. Large doses of ionizing radiation cause in animals so called (post) radiation disease. Its cause is:

- a) disruption of cell membranes;
- b) damage to the enzymes catalysing basic metabolic processes;
- c) killing of symbiotic microorganisms in the alimentary tract, and consequently inadequate supply of some vitamins;
- d) damage to DNA, interfering with cell multiplication;
- e) always answers is correct.

104. Graph illustrates the speed of enzyme-catalysed reaction (V) as a function of temperature (temp). Why at temperatures t_1 and t_2 the reaction speeds are the same?

- a) because all chemical reactions have an optimum temperature, above and below they progress at lower speeds;
- b) at t_1 temperature only a few enzyme molecules are active, but they form very stable complexes with their substrate molecules, whereas at t_2 the opposite is true;
- c) at t_2 less enzyme molecules are active, than at t_1 , but each of these molecules is capable of converting more substrate molecules per time unit than at t_1 ;
- d) at both temps. the number of active enzyme molecules is the same, but at t_1 each of them forms more stable enzyme-substrate complexes than at t_2 , and t_2 each enzyme molecule works faster, than at t_1 ;
- e) all the above explanations are incorrect.

105. Among known extant (contemporary) animal species, invertebrate species constitute about:

- a) several %;
- b) a dozen % (between 10 and 20);
- c) roughly 50%;
- d) roughly 75%;
- e) above 90%.

106. A group of plants as "algae" is characterized jointly (especially in older textbooks), because:

- a) they are a true monophyletic group;
- b) they are polyphyletic, but nevertheless show remarkable similarity their physiology and chemical constitution (photo - synthetic pigments, storage material etc);
- c) they are polyphyletic, but nevertheless have almost identical life cycles;
- d) they are polyphyletic, but nevertheless exhibit similar ecology and degree of morphological complexity;
- e) more than one the above explanations is correct.

107. Mimicry is the result of evolutionary process when organisms of one species mimic the features (usually bright coloration) of another species (the "master specie"). USUALLY the organisms of the "master species" are:

- a) very closely related to the mimicking species (they form so called sibling species);
- b) much larger, than those of the mimicking species;
- c) poisonous or inedible;
- d) innocuous, but the organisms of the mimicking species are poisonous;
- e) more than one of the above answers is correct.

108. " Fight for survival" is a consequence of:

- a) innate tendency to compete;
- b) the necessity to fight against natural disasters (floods, volcanic eruptions, temperature extremes etc);
- c) genetic variability;
- d) the number of offspring organisms exceeding capacity of their habitat;
- e) all of the above.

109. Palinology is the science of fossilized pollen. The best preserved pollen grains are likely to be found in:

- a) crude oil;
- b) peat;
- c) coal;
- d) limestone;
- e) sandstone.

110. Some moss (bryophyta) and club-moss (Lycopodiophyta) species may be

morphologically quite alike, but they differ by numerous characters.

Which of the following is correct:

- a) if in the transverse section of the stalk one observes specialized water-conducting cells - the species certainly belongs to club - mosses;
- b) if in the cells of the "leaves" one sees an odd number of chromosomes - the species certainly belongs to mosses;
- c) if all spores of the organism are identical - the species certainly belongs to mosses;
- d) all of the above are correct;
- e) neither of the above is correct.

111. In the blood of an adult man the total content of haemoglobin is, roughly:

- a) several hundred gram;
- b) tens of gram (10-100 g);
- c) several gram;
- d) several hundred milligram;
- e) tens of milligram.

112. The action of a chemical inhibiting the enzyme acetylcholinesterase will cause, as a primary response:

- a) complete relaxation of all muscles;
- b) relaxation only of skeletal (striated) muscles;
- c) contraction of most of the muscles. both striated and smooth;
- d) contraction of some of the striated muscles only;
- e) contraction of some smooth muscles only.

.113. Penicillin does not harm nondividing bacterial cells, but in its presence bacteria cannot increase in number. From this information it is reasonable to assume that penicillin inhibits:

- a) transcription;
- b) protein biosynthesis;
- c) biosynthesis of cell-wall component(s);
- d) DNA replication;
- e) more than one of the above is consistent with the given information.

114. Callus cells were incubated for several hours in a medium containing all necessary substances, with one of the chemicals being radioactive(tritium-labelled). Afterwards the cells were fixed for microscopy; using autoradiography it was observed that the radioactivity was concentrated exclusively in the nucleus,the mitochondria and chloroplasts. It is reasonable to assume that the compound labeled was?

- a) an aminoacid;
- b) uridine;
- c) thymidine;
- d) glucose;
- e) more than of the above possibilities is very likely.

115. Rough endoplasmic reticulum should be particularly abundant in:

- a) reticulocytes (immature erythrocytes);
- b) nerve cells;
- c) pancreas cells;
- d) smooth muscle cells;
- e) it is abundant to a similar degree in all of the above types of cells.

116. Correct ordering of structures, from the least complex one (that is, containing the lowest number of different kinds of protein molecules) to the most complex (that is, containing the largest number of different kinds of protein molecules) is given in answer:

- a) scrapie prion; HIV virion; eukaryotic ribosome; mitochondrion;
- b) eukaryotic ribosome; HIV virion; scrapie prion; mitochondrion;
- c) scrapie prion; HIV virion; mitochondrion; eukaryotic ribosome;
- d) HIV virion; scrapie prion; eukaryotic ribosome; mitochondrion;
- e) scrapie prion; eukaryotic ribosome; HIV virion; mitochondrion.

117. Polyribosomes may be found free in the cytoplasm (cytosol), or attached to the endoplasmic reticulum. Which factor determines whether a molecule of mRNA is translated in "free" or in "bound" polyribosomes?

- a) it depends on the amino acid sequence of the polypeptide coded for by this mRNA;
- b) if the mRNA has a poly(A)-tail - it will be translated in bound polysomes, if it has no tail - it will be translated in free polyribosomes;
- c) if the mRNA has a 5' cap - it will be translated in bound polysomes, if it has no cap - it will be translated in free polyribosomes;
- d) only mRNAs shorter than 1000 nucleotides are translated in free polyribosomes; the longer ones will be translated in bound polyribosomes;
- e) in fact it depends on the type of the cell - in certain cells mRNA get translated mostly in free polyribosomes, in other types of cell - mostly in bound polyribosomes;

118. The substances given below are all coloured. For which of them this fact has no direct bearing on its functioning?

- a) chlorophyll;
- b) phytochrome;
- c) vitamin A;
- d) haemoglobin;
- e) for all the above substances their colour is essential for their functioning.

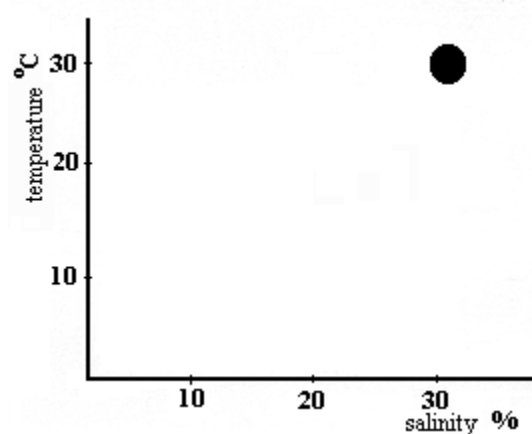
119. Ecological assemblies K through Q consist of species designated with numbers 1 through 8, present at various densities. Individual densities of these species in any particular assembly are given (as individuals per square meter) in the table. Which of the above assemblies is the least susceptible to a massive pest infection (gradation):

assembly	K	L	M	N	P	Q
species 1	50	92	75	0	0	0
species 2	30	4	5	25	2	65
species 3	10	0	5	20	3	20
species 4	10	0	5	20	5	10
species 5	0	1	5	20	40	3
species 6	0	1	5	5	50	2
species 7	0	1	0	0	0	0
species 8	0	1	0	0	0	0

- a) K;
- b) L;
- c) M;

- d) N;
- e) P.

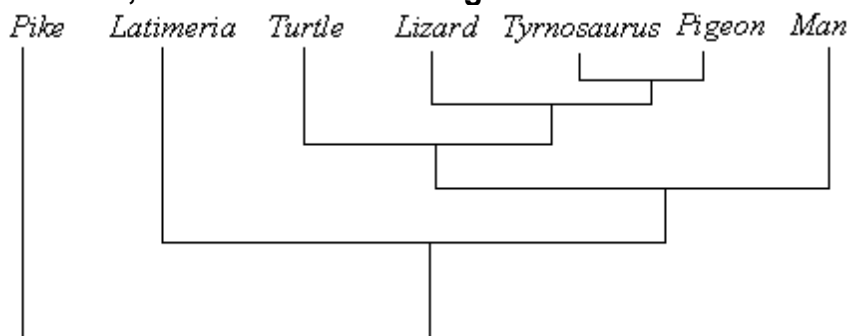
120. Tolerance spectra of water temperature and salinity of a crustacean species is shown in the graph. This species is likely to be found on New Year's eve (Christian) in surface waters of:



- a) an alpine pond;
- b) Aral Sea;
- c) Indian Ocean;
- d) The Dead Sea;
- e) in more than one of the above.

121. A simplified genealogy tree of seven species is shown. Based solely

on this tree, which of the following statements is correct:



- a) the evolutionary relationship of tyrannosaurus and pigeon is closer than of tyrannosaurus and lizard;
- b) turtle is more closely related to lizard than to tyrannosaurus;
- c) man is more closely related to pigeon than to turtle;
- d) two of the above statements are correct;
- e) statements A., B. and C. are correct.

122. In the Earth's past there were several episodes of sudden (in the geological time-scale) extinction of a significant number of species. During one such period dinosaurs, among other groups, became extinct. This took place:

- a) several million years ago (Myr);
- b) a dozen 10-20) Myr;
- c) somewhat more than 50 Myr;
- d) ca. 100 Myr;
- e) more than 200 Myr.

123. Animal behaviour patterns, in which an individual endangers its life to benefit other members of the group, are called altruistic. It is believed that altruistic behaviour was favoured by kin selection. Which of the examples given below CANNOT be explained as kin-selection-favoured?

- a) suicidal attack by a worker bee guarding its hive;
- b) protection of the queen of an ant species by "soldier ants";
- c) protection of lion cubs by a lioness NOT being their mother;
- d) warning cries of a bird warning other individuals about approaching danger;
- e) in fact, all of the above behaviour patterns probably arose through kin selection.

124. School forming by fish, herd forming by herbivores, aggregation forming by planctonic organisms:

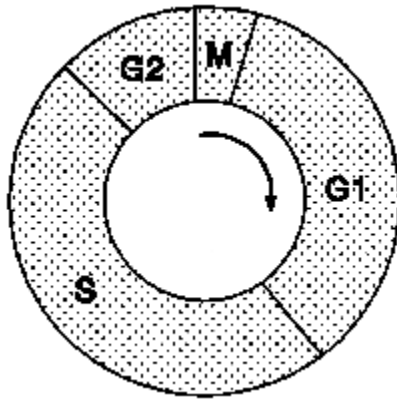
- a) reflects the organisms' intrinsic social needs;
- b) makes it easier to obtain food;
- c) protects individual organisms against a predator attack;
- d) helps protect the progeny;
- e) is only temporary and solely reflects local abundance food.

125. A retrovirus is causing AIDS, the so called HIV-virus. A retrovirus contains RNA, which the host cell is transcribed in DNA, with the help of the enzyme reverse-transcriptase. Which of the following statements is correct?

- I. Viral DNA is synthesized in cells infected with the HIV-virus;
- II. Viral RNA is synthesized in cells infected with the HIV-virus;
- III. DNA is translated to viral protein.

- a) I;
- b) II;
- c) III;
- d) I, II;
- e) I, II, III.

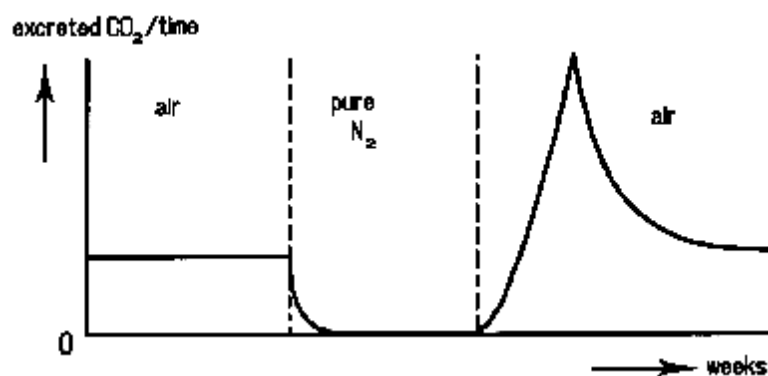
126 The picture shows a schematic drawing of the cell cycle.



Somebody wants to determine the duration of the S-phase. This is done in adding with tritium labeled compound R to the medium of the organism with the dividing cells. Which of the following compounds is the most suitable to be R?

- a) adenine;
- b) cytosine;
- c) guanine;
- d) thymine;
- e) ATP;

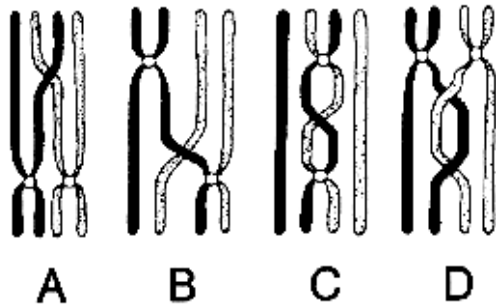
127. Potatoes are stored during one week in pure air, after this during one week in pure nitrogen and in pure air again. During the experiment the excretion of CO_2 is measured. The diagram shows the results.



The extra amount of CO_2 being produced and excreted during the third week probably originates from:

- a) ethanol;
- b) ethanal;
- c) lactic acid;
- d) NAD H_2 ;

128. Four pupils have the following schematic drawings of chromosomes during Meiosis I.



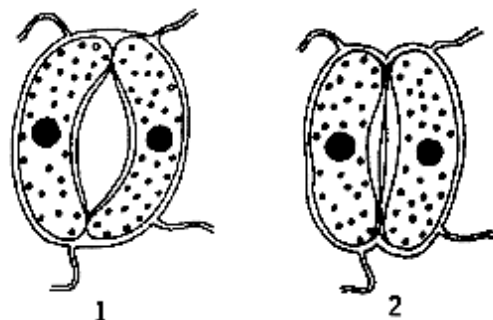
In which drawing(s) the position of the chromosomes allow the appearance of crossing over?

- a) A, B;
- b) A, D;
- c) A, C;
- d) B, D;
- e) D, C.

129. A certain plant doesn't take up radio active substances from the soil unless fungi are present in and between the root cells. Four statements derived from this information are stated below. These fungi do influents:

- a) the intake of substances by root hairs;
- b) The protection of the root by the rootcap;
- c) watertransport by xylem;
- d) the growth of the root by meristem;

130. Examine the two drawings of the same stoma different moments.



In which drawing turgor is highest in the guard cell?

In which drawing the concentration of dissolved particles is highest?

- | | Turgor highest | concentration highest |
|----|----------------|-----------------------|
| a) | 1 | 1 |
| b) | 1 | 2 |
| c) | 2 | 1 |
| d) | 2 | 2 |

131. During one hour in light condition the evaporation is measured of three different but equal sized plants. After this, just above the ground

the stem is cut off and during one hour the amount of released fluid is measured.

Results:

	evaporation	released fluid
chinese rose	6,20 ml	0,02 ml
sunflower	4,80 ml	0,02 ml
tomato	0,50 ml	1 0,07 ml

John concludes:

root pressure has a considerable influence on the watertransport

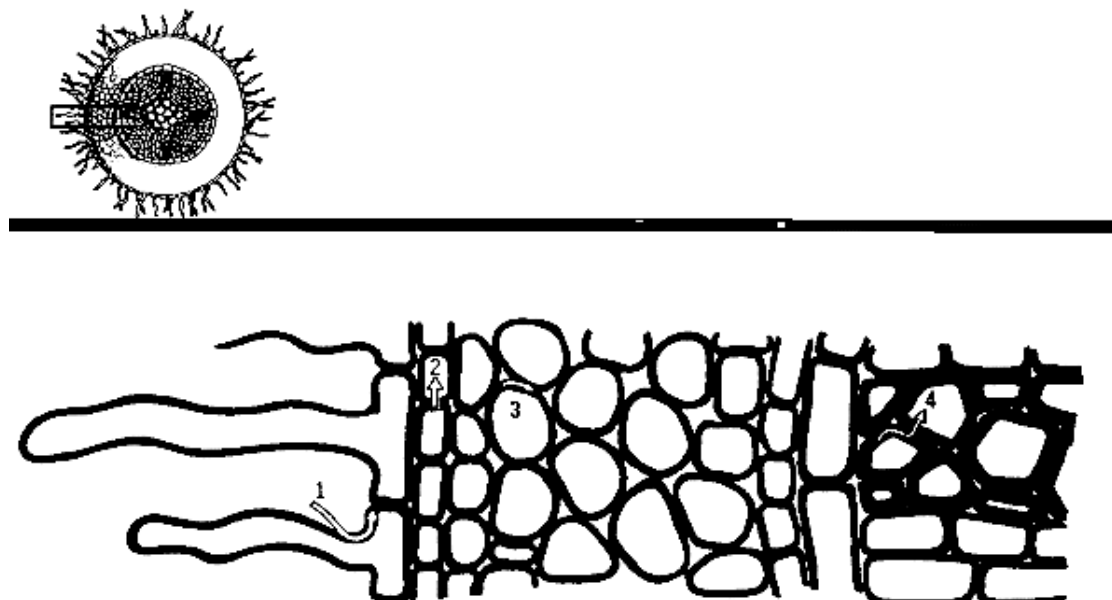
Mary concludes:

capillary forces have a considerable influence on the watertransport.

Who is right?

- a) John and Mary;
- b) only John;
- c) only Mary;
- d) neither John,nor Mary;+

132 Examine the figure showing a drawing + magnification of a piece of a cross section of a root of a dicotyledon plant.



Cell walls are indicated in black.

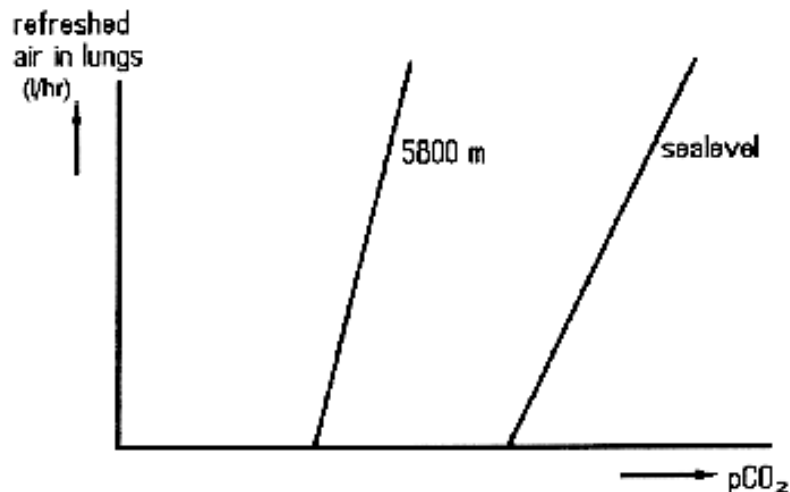
Four arrows are visible in the drawing.

Indicate which arrow(s) could represent active transport of and ion.

- a) 1;
- b) 2;
- c) 3;

d) 4;

133 The influence of $p\text{CO}_2$ on the amount of refreshed air in the lungs is determined of a test person. The experiment is done at sealevel and high up in mountains at 5800m. The diagram shows the results.



Examine the two following statements:
 the sensitivity of the respiratory centre for CO_2 increases if $p\text{O}_2$ decreases.
 the influence of $p\text{CO}_2$ on the amount of refreshed air in the lungs increases if $p\text{O}_2$ decreases. What is correct?

- a) none of these;
- b) only I;
- c) only II;
- d) both I and II;

134. Inspect the following table with data

	total concentration mmol/l	concentration of Na^+ mmol/l)
I. blood of fish I (sea fish)	1050	290
II.. blood of fish II (sea fish)	360	190
III. blood of fish III (fresh water fish)	280	130
seawater	1000	470
fresh water	10	1

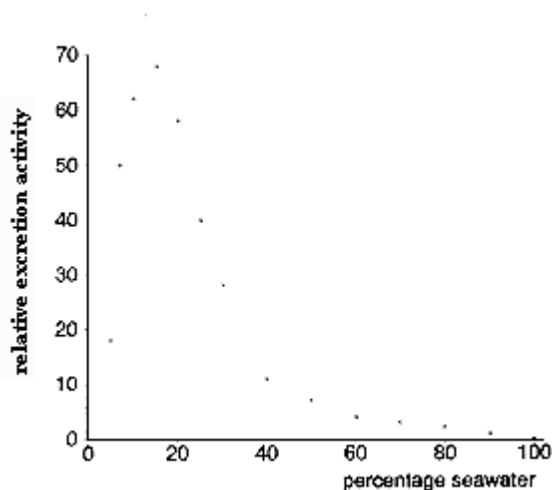
Diffusion of water and Na^+ may occur in the gills of these fishes.
 Which fish(es) will have for both water and Na^+ :

excretion / intake < 1

- a) I;
- b) II;

c) III.

135 The relative excretion activity is determined of the protozoan *Zoothamnium* in different mixtures of seawater and fresh water. The results are shown in the diagram.



Consider the following conditions:

I. below 15 % seawater

II. 15 seawater

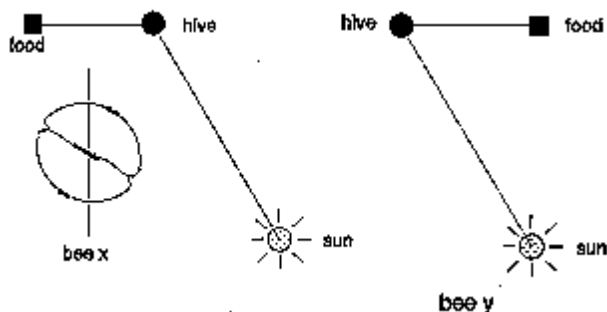
III. 15-40 % seawater

IV. above 40% seawater

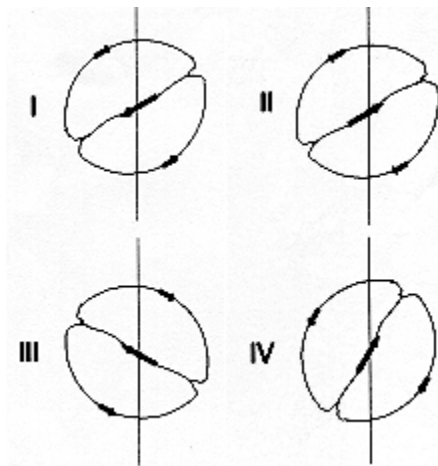
Which of these four conditions is most suitable for this protozoan?

- a) I;
- b) II;
- c) III;
- d) IV.

136 A bee X has found a location with pollen supplies. Her waggle dance in a hive is shown in the drawing. Another bee Y also has found a location with food.

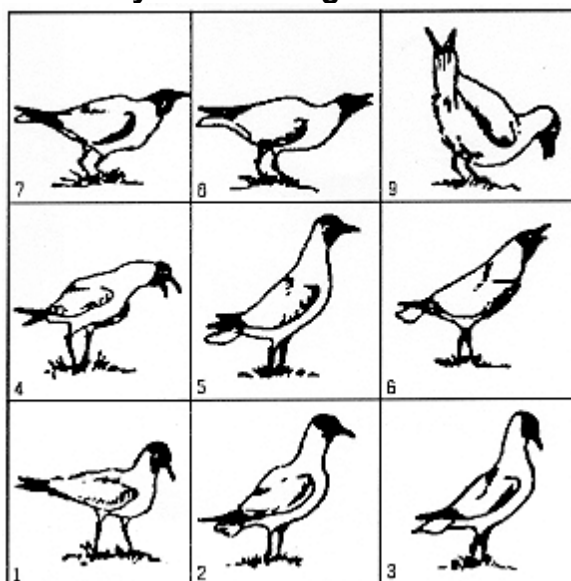


Which of the following drawings represents the waggle dance of bee Y?



- a) I;
- b) II;
- c) III;
- d) IV;

137 The figure shows possible positions of the black headed gull (*Larus lanoe phalus*).A gull has landed by accident in the territory of another gull and both gulls have met each other. Which position will probably be shown by the visitor gull?



- a) position 1;
- b) position 3;
- c) position 5;
- d) position 7;
- e) position 9;

138 In a certain human population 64% is able to roll the tongue. This ability is based on a dominant allele. A roller marries a non-roller. Calculate the chance to have a roller baby.

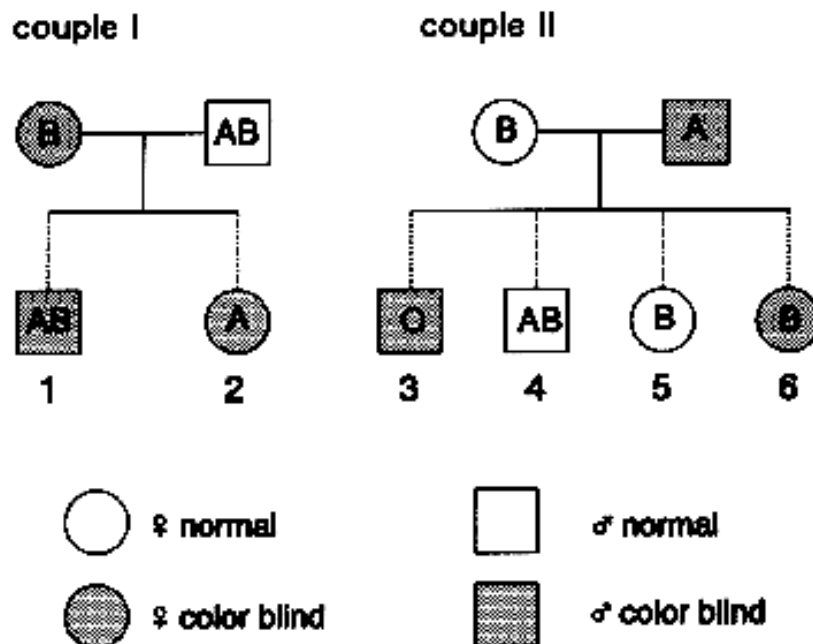
- a) 0,1;
- b) 0,225;

- c) 0,5;
- d) 0,625;
- e) 0,75.

139. A certain feature of a plant is based on three alleles: E1, E2 and E3. A pollen grain of this plant is unable to create a pollen tube if one and the same allele exists in both the pollen grain and the cell of the pistil. A plant with genotype E1E2 is pollinated with pollen of a plant with genotype E1E3. Which genotype(s) is (are) to be expected in the offspring?

- a) E1E1 E1E2;
- b) E1E3 E2E3;
- c) E1E3 E1E2.

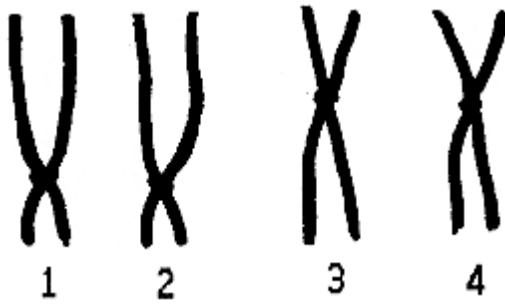
140 Three alleles are determining the ABO-bloodgroups: I^A , I^B and i . Color blindness is caused by a recessive X-chromosomal allele. Inspect the following family trees.



As you can see some of the parents and children are colorblind. Also the bloodgroups are indicated. Directly after birth a baby of couple I must have been changed with a baby of couple II. Indicate the numbers of the changed babies.

- a) 1 - 3;
- b) 2 - 6;
- c) 2 - 5;
- d) 2 - 4;
- e) 1 - 5.

141 The figure shows a karyogram of a cell of an individual with $2n = 4$.



View the following statements:

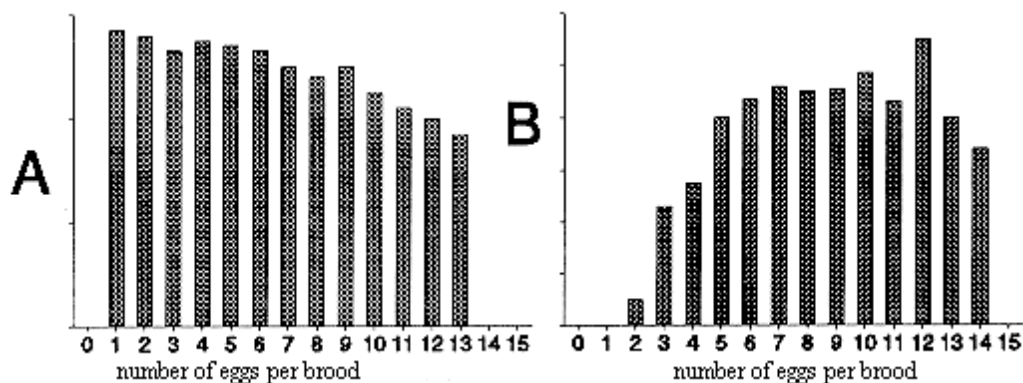
- I. If chromosome 1 is from the father, then chromosome 2 is from the mother;
- II. Chromosomes 1 and 2 are from the father and 3 and 4 are from the mother;
- III. During meiosis I chromosomes 1 and 2 go to one pole while, 3 and 4 are going to another pole;
- IV. During meiosis I chromosomes 1 and 3 go to one pole while 2 and 4 are going to another one. Correct are:

- a) I and III;
- b) I and IV;
- c) II and III;
- d) II and IV;

142. Someone has investigated the fitness of a population of the coletit (Parus major) in relation to the number of eggs per nest. He determined:

- I. The frequency of number of eggs per nest;
- II. The average weight of the young birds at the moment they flew out;
- III. The number of birds caught back per nest at the end of the summer;
- IV. The average area of a territory;

The two diagrams A and B are showing the results of two of these four aspects.

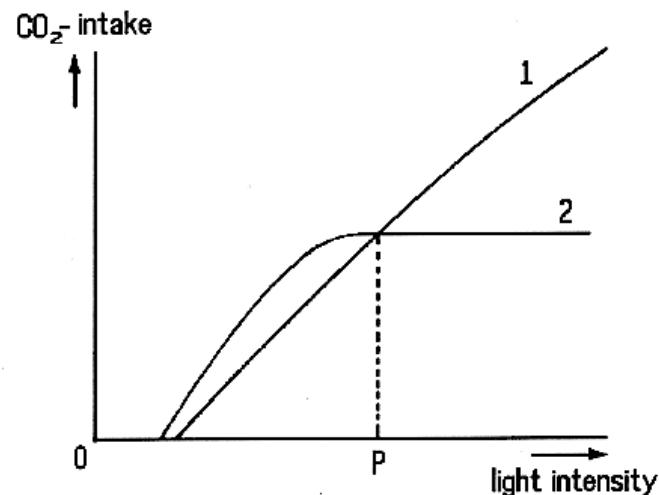


What are the diagrams representing?

- | | |
|-------|------|
| A | B |
| a) I | II; |
| b) II | III; |

- c) I IV;
d) II IV.

143. The diagram shows the relationship between the intake of CO_2 and light intensity of leaves of a beech tree in a sunny and in a shadow position.



Examine the two statements:

- I. graph 1 matches with leaves in the shadow position
II. light intensity is a limiting factor for shadow leaves at the situation marked with letter P.

What is correct?

- a) both I and II;
b) only I;
c) only II;
d) none of these;

144. How many kinds of gametes can an individual with n pairs of chromosomes produce? (assuming no exchange of part between homologous chromosomes.).

- a) 2^{n-1} ;
b) n^2 ;
c) n ;
d) 2^n ;
e) 2.

145. From the of being hatched from the egg, a nestling duck was raised fully by hand, he or she will think of the raiser as his (her) parents, the behavioral mechanism of this phenomena is:

- a) learning behavior;
b) imprinting;
c) adaptation strategy;
d) recognition.

146. When a new male, takes over a lion pride, they sometimes kill or evict

the cubs already present, this phenomena can be explained from behavioral ecology:

- a) the male doesn't like cubs;
- b) the male cannot afford too much for caring those cubs;
- c) the male breed his own offspring;
- d) degeneration of the male's parental behavior.

147. Honey-bee is a kind of social insects. In its colony, males develop from

unfertilized eggs and are haploid, while workers develop from normal fertilized

eggs and are therefore dipliid. Which of the following is the correct coefficient of realatadness (R) between workers on the colony:

- a) 0.5;
- b) 0.25;
- c) 1;
- d) 0.75;
- e) 0.375.

148. There is significant correlation between intensity of sexual selection and mating system. Which of the following is correct?

- a) intensity of sexual selection in promiscuity is strong;
- b) intensity of sexual selection in monogamy is strong;
- c) intensity of sexual selection in polygyny is strong;
- d) intensity of sexual selection in polyandry is strong.

149. Which is the determinant factor in distinguishing dominant from subordinate for animals living in group?

- a) aggressive;
- b) amicable;
- c) territory;
- d) recognition.

150. In addition to providing immediate energy, which of the followings is the most prominent contribution of glycolysis to aerobic respiration?

- a) glucose production;
- b) NAD + production;
- c) ATP production;
- d) lactic acid production;
- e) pyruvate production.

151. Which of the following is the rate-limiting step in glycolysis?

- a) isomerization of glucose-6-phosphate to fructose-6-phosphate;
- b) phosphorylation of fructose-6-phosphate;
- c) phosphorylation of glucose
- d) conversion of pyruvate to lactate.

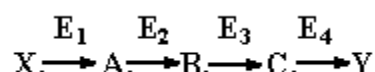
152. Which of the followings is observed in an experimental animal fed on a diet composed of excessive fat and insufficient carbohydrate?

- a) a decrease in acetyl-CoA synthesis;
- b) an extremely high consumption of glucose;
- c) a decrease in fat metabolism;
- d) ketosis;
- e) conversion of fatty acids to glucose.

153. Which of the following substances can pass through the cell membrane only by means of endocytosis and exocytosis?

- a) proteins, amino acids, monosaccharides;
- b) oil droplets, gases, fatty acids;
- c) proteins, oil droplets, polysaccharides;
- d) oil droplets, water, ions;
- e) polysaccharides, monosaccharides, water.

154. In a synthetic chain reaction:



if the enzyme E4 has a defect, in this case, at least which of the followings has to be added to the medium in order the bacteria to grow normally through autotrophic way?

- a) X;
- b) A;
- c) X and C;
- d) C;
- e) Y;

155. The following table shows the percentages of nucleic acid bases, isolated from different species.

Species	Adenin	Guanin	Thymine	Cytosine	Uracil
1	21	29	21	29	0
2	29	21	29	21	0
3	21	21	29	29	0
4	21	29	0	29	21
5	21	29	0	21	29

According to these information. Which of the following statements is false?

- a) species (1), (2) and (3) carry DNA as the genetic material;
- b) species (4) and (5) carry RNA as the genetic material;
- c) species (1) and (2) have double-stranded DNA;
- d) species (3) has single-stranded DNA;
- e) species (4) and (5) have double-stranded and single-stranded RNA respectively.

156. In an intermediary heredity, which of the following crosses gives rise to offsprings with only two different phenotype?

- a) $TT \times TT$;
- b) $Tt \times Tt$;
- c) $Tt \times tt$;
- d) $TT \times tt$;
- e) $tt \times tt$;

157. Two organisms with a genotype of $TtGg$ (T=height, G=color) are mated with each other. Which of the followings the probability for the offsprings to carry only one of the dominant characters?

- a) $9/16$;
- b) $7/16$;
- c) $6/16$;
- d) $3/16$;
- e) $15/16$;

158. Eucaryotic genes may not function properly when cloned into bacteria.

Which of the followings is not a reason for this?

- a) inability to excise introns;
- b) destruction by native endonucleases;
- c) failure of promoter to be recognised by bacterial RNA polymerase;
- d) different ribosome binding sites;
- e) using different genetic codes.

159. Suppose that you are trying to clone a specific human gene. After human DNA fragments have been spliced into plasmid vectors (shotgun method) and the plasmids have been exposed to recipient bacterial cells, which of the following kind of bacteria you would not find?

- a) bacteria that contain free spliced human DNA fragments;
- b) bacteria containing plasmid with no human DNA in it;
- c) bacteria that contain plasmid DNA with the desired human gene in it;
- d) bacteria containing plasmid with irrelevant human gene in it;
- e) bacteria that do not contain any plasmid;

160. While the stem of a high plant is branching, if the main stem develops

better than the branches and becomes dominant to them, which of the following names is given to such a branching?

- a) monocasium;
- b) monopodial;
- c) dicasium;
- d) sympodial;
- e) pleiocasium.

161. If a plant is allowed to make photosynthesis in a CO_2 containing medium, to which of the following molecules does the radioactive carbon

incorporate first?

- a) PGA;
- b) phosphoglyceraldehyde;
- c) NADPH;
- d) ribulose-1,5 diP;
- e) pyruvic acid.

162. Which of the following *Cyanophyta* member cannot fix the atmospheric nitrogen?

- I. *Anabaena*;
- II. *Scenedesmus*;
- III. *Calotrix*;
- IV. *Nostoc*.

- a) I, IV;
- b) III, IV;
- c) I, III;
- d) I, III, IV;
- e) only II;

163. A mollusc sample is given to a biologist. After examining the sample he says that it belongs to *Bivalvia*. Which of the followings may be the key that makes him to reach this conclusion?

- a) gills;
- b) absence of radula;
- c) body symmetry;
- d) mantle.

164. Which of the followings occurs if an acetylcholine esterase inhibitor is added to a synaptic cleft?

- a) postsynaptic neuron is continuously stimulated;
- b) postsynaptic neuron cannot be stimulated;
- c) presynaptic neuron does not release neurotransmitters;
- d) presynaptic neuron cannot be stimulated;
- e) postsynaptic neuron membrane is hyperpolarised.

165. Which of the following impulse conducting ways is activated if our head makes rotational movements?

- a) utricle and saccule—cerebellum—brain cortex;
- b) semicircular canals--cerebellum--brain cortex;
- c) semicircular canals--brain cortex--cerebellum;
- d) utricle and saccule--brain cortex—cerebellum.

166. An extremely high amount of ammonia is found in a soil sample. Which of the followings is the reason of this?

- a) absence of nitrogen fixing fungi;
- b) absence of bacteria which free the bound nitrogen;
- c) competition among bacteria;
- d) absence of nitrogen fixing bacteria;
- e) inability of rocks to make absorption.

167. With which of the followings, convergent evolution is brought about between two species?

- a) occurrence of similar mutations;
- b) occurrence of hybridization;
- c) exposure to similar selection stresses;
- d) hybridization of each of the two species with a third one;
- e) genetic drift between two species.

168. Choose the correct answer. Some of the cell components are:

- 1. Cytoplasm;
- 2. Mitochondrions;
- 3. Ribosomes;Chromatine;
- 4. Cell wall;
- 5. Nucleus;
- 6. Centriole.

All the procariotic cells contain:

- a) 1.,2., 4., 5.
- b) 1., 2., 4., 6.
- c) 1.,2., 3., 5., 7.
- d) 1., 3., 4., 5.

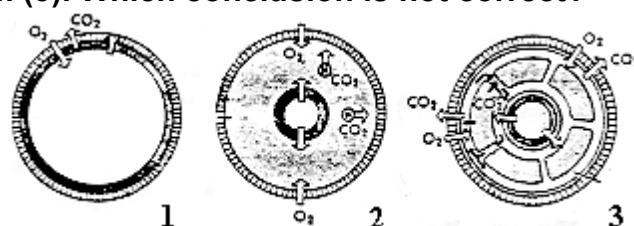
169. The group of animals, developing by the change of the host, is:

- a) *Hirudinea, Ixodes, Ostrea*
- b) *Astacus, Pediculus, Aphis*
- c) *Ichneumon, Ascidia, Ascaris*
- d) *Plasmodium, Fasciola, Trichinella*

170. Hydrogen peroxide is dropped out by drops on the slices of boiled and unboiled potatoes. Foam appears only on slice X, but it does not appear on slice Y. Which is the correct explanation?

- a) slice x is boiled, because hydrogen peroxide affects only carbohydrates decomposed in monosacharides.
- b) slice x is boiled, because ferments in unboiled potatoes make hydrogen peroxide inactive.
- c) slice x is unboiled, because there are ferments in the plants, promoting the decomposition of hydrogen peroxide, but they denaturate in high temperatures.
- d) slice x is unboiled, because hyrogen peroxide decomposes ferments in it.

171. You can see cross - section diagrams of hydra (1), planaria (2) and earth- worm (3). Which conclusion is not correct?



- a) those animals have no special respiration organs.

- b) hydra and earth - worm have body cavity, but planaria do not .
- c) earth - worm has circulatory system, but hydra and planaria do not .
- d) all those animals are aerobic organisms.

172. Phenylketonuria is a recessive human genetic disorder, causing the disturbance of aminoacids metabolism. Phenylketonuria causes :

- a) inability to synthesize phenylalanine.
- b) inability to decompose phenylalanine.
- c) inability to absorb phenylalanine.
- d) inability to include phenylalanine in protein synthesis.

173. In which cases the transpiration in plants intensifies ?

- 1. The air temperature and the rate of wind increases.
- 2. The light intensity weakens and the weather becomes calm .
- 3. The water amount in soil and the air temperature decreases.
- 4. The air temperature decreases and the night begins.

- a) 1 and 3
- b) 1 only
- c) 4 only
- d) 2 only

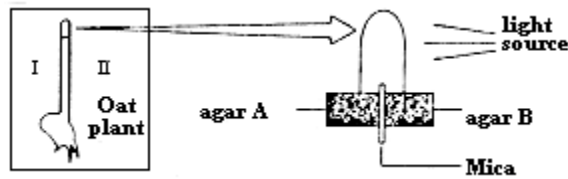
174. Look at the pictures illustrating the structure of fresh water fish nephron (1) and marine fish nephron (2). Which one of the statements is correct ?



- a) marine fish loses more water with urine than fresh water fish.
- b) the absence of glomeruli helps marine fish to get rid of spare water.
- c) salts reabsorb more intensively in marine fish tubules, than in fresh water coiled tubules.
- d) glomeruli help fresh water fish to get rid of spare water.

*****Following information deals with the questions №175,176**

The next items are based on the following diagram illustrating a coleoptile tip from a plant grown in the dark and placed on blocks of agar separated by a sheet of mica.



175. The tip is exposed to light as shown. The influence effect of the light on the distribution of auxin will be such as:

- a) block A will contain more than block B.
- b) block B will contain more than block A.
- c) both blocks will contain the same amount.
- d) neither block will contain auxin.

176. After exposure to light as shown, agar blocks alone are placed back on the cut surface on the oat plant so that a half of the (side I) is covered by block A and an other half (side II) is covered by block B. It is likely that

- a) neither side will grow.
- b) both sides will grow equally fast.
- c) side II will grow faster than side I.
- d) side I will grow faster than side II.

*****Following information deals with the questions №177-180:**

The codes of the answers:

- 1. Body is divided into carapace and abdomen;
- 2. Head, chest, abdomen carapace - abdomen;
- 3. Two pairs of antenna;
- 4. No antennas;
- 5. One pair of oral members;
- 6. The stage of larva is presented;
- 7. No larval forms;
- 8. Three pairs of ambulatory legs;
- 9. Four pairs of ambulatory legs;
- 10. Imagoes breath only through tracheas;

177. Choose the features typical for representatives of the class Crustacea.

- a) 5, 7, 10;
- b) 2, 6, 8, 10;
- c) 1, 3, 6;
- d) 4, 7, 9.

178. Choose the features typical for the representatives of the class Myriapoda

- a) 5, 7, 10;
- b) 2, 6, 8, 10;
- c) 1, 3, 6;

d) 4, 7, 9.

179. Choose the features typical for representatives of the class *Insecta*.

- a) 5, 7, 10;
- b) 2, 6, 8, 10;
- c) 1, 3, 6;
- d) 4, 7, 9.

180. Choose the features typical for representatives of the class *Arachnida*.

- a) 5, 7, 10;
- b) 2, 6, 8, 10;
- c) 1, 3, 6;
- d) 4, 7, 9.

*****Following information deals with the questions №181-183:**

The code of the answers:

- 1. Have highly - developed conductive tissues;
- 2. Gametophyte prevail;
- 3. Sporophyte prevail;
- 4. Male gametes - spermatozoides;
- 5. Male gametes - spermatozoid
- 6. Equal - spored;
- 7. Hetero - spored;
- 8. Equal - spored or hetero - spored;
- 9. Have roots;
- 10. Haven't roots.

181. Chose the features which are typical for *Pteridophyta*.

- a) 1, 3, 5, 7, 9;
- b) 1, 3, 4, 8, 9;
- c) 2, 4, 6, 10.

182. Chose the features which are typical for *Bryophyta*.

- a) 1, 3, 5, 7, 9;
- b) 1, 3, 4, 8, 9;
- c) 2, 4, 6, 10.

183. Chose the features which are typical for *Spermatophyta*.

- a) 1, 3, 5, 7, 9;
- b) 1, 3, 4, 8, 9;
- c) 2, 4, 6, 10.

*****Following information deals with the questions №184,185:**

The code of the answers:

- 1. Have only pulmonary breathing;
- 2. Poikilotherma amniotes;
- 3. Have urinary bladder;

4. *Secrete the products of nitrogenous metabolism chiefly in a form of acid;*
5. *Secrete the products of nitrogenous metabolism chiefly in a form of urea;*
6. *Have mesonephric kidneys;*
7. *Molting is typical for imagoes;*
8. *Larva stage is present;*
9. *Three - chambered heart;*

184. Using the code denote which features are typical for *Amphibia*

- a) 2, 3, 6, 7, 9;
- b) 3, 5, 6, 8, 9;
- c) 1, 2, 3, 4, 7, 9;
- d) 1, 4, 5, 6, 8, 9;

185. Using the code denote which features are typical for *Reptilia*.

- a) 2, 3, 6, 7, 9;
- b) 3, 5, 6, 8, 9;
- c) 1, 2, 3, 4, 7, 9;
- d) 1, 4, 5, 6, 8, 9;

*****Following information deals with the questions №186-188;**

Codes of the answers:

1. Trees bushes, herbaceous plants;
2. Trees, bushes;
3. Trees, herbaceous plants;
4. Haploid endosperm (n);
5. Diploid endosperm (2n);
6. Triploid endosperm (3n);
7. Male gametes - spermatozoons;
8. Male gametes - spermatozooids;
9. Male gametes - spermatozoons or sperma;
10. Insemination isn't connected with water;
11. Insemination is connected with water;
12. Insemination of ones is connected with water, while of others is not;
13. Vegetative organs have vessels;
14. Vegetative organs have not vessels;
15. Have specialized vegetative organs (tubers, rhizomes, bulbs).

186. Chose the features typical for *Angiospermatophyta*

- a) 1, 6, 8, 11, 13, 15;
- b) 3, 8, 11, 13, 15;
- c) 2, 4, 9, 12, 13;
- d) 1, 6, 7, 10, 13, 15;
- e) 2, 3, 8, 12, 15.

187. Chose the features typical for *Gymnospermatophyta*

- a) 1, 6, 8, 11, 13, 15;
- b) 3, 8, 11, 13, 15;
- c) 2, 4, 9, 12, 13;
- d) 1, 6, 7, 10, 13, 15;

e) 2, 3, 8, 12, 15.

188. Chose the features typical for *Pteridophytes*

- a) 1, 6, 8, 11, 13, 15;
- b) 3, 8, 11, 13, 15;
- c) 2, 4, 9, 12, 13;
- d) 1, 6, 7, 10, 13, 15;
- e) 2, 3, 8, 12, 15.

*****Following information deals with the questions №189-191;**

The code of the answers:

- 1. *Autotroph;*
- 2. *Heterotroph;*
- 3. *They have chlorophyllin protoplasm;*
- 4. *They have chloroplasts;*
- 5. *They have mitochondrions;*
- 6. *They have a nucleus;*
- 7. *Genetic material is situated in cytoplasm;*
- 8. *They have cytoplasmic membrane;*
- 9. *They have cellulose membrane;*
- 10. *They can move;*
- 11. *They can't move.*

189. Denote, which processes of vital activity and peculiarities of structure are characteristic for cells of animals.

- a) 2, 5, 6, 8, 10;
- b) 1, 4, 5, 6, 8, 9, 11;
- c) 2, 5, 6, 8, 9, 11;
- d) 1, 2, 7, 8, 9, 10, 11;
- e) 1, 3, 7, 8, 9, 10, 11.

190. Denote, which processes of vital activity and peculiarities of structure are characteristic for cells of fungi

- a) 2, 5, 6, 8, 10;
- b) 1, 4, 5, 6, 8, 9, 11;
- c) 2, 5, 6, 8, 9, 11;
- d) 1, 2, 7, 8, 9, 10, 11;
- e) 1, 3, 7, 8, 9, 10, 11.

191. Denote, which processes of vital activity and peculiarities of structure are characteristic for cells of bacterium.

- a) 2, 5, 6, 8, 10;
- b) 1, 4, 5, 6, 8, 9, 11;
- c) 2, 5, 6, 8, 9, 11;
- d) 1, 2, 7, 8, 9, 10, 11;
- e) 1, 3, 7, 8, 9, 10, 11.

*****Following information deals with the questions №192,193;**

The code of the answers:

1. Diameter 25 nm;
2. Diameter 8-10 nm;
3. Diameter 7 nm;
4. Tubular;
5. Consists of various proteins, e.g. karatine;
6. Tubes consist of 4-5 protofilaments;
7. Consists of molecules of tubuline, which is a protein;
8. Consists mainly of actin;
9. Frms horn substance death of cells;
10. Eementary units from which centrioles, nucleous spindles, flagella and cilia.

192. The eucaryotic cell contains filaments and tubular structures which are termed cytoskeleton. Please match the microtubuli with the corresponding statements numbered above

- a) 1, 4, 7, 10;
- b) 3, 6, 8;
- c) 2, 5, 6, 9;
- d) 3, 8.

193. Please match the microfilaments with the corresponding statements as numbered above.

- a) 1, 4, 7, 10;
- b) 3, 6, 8;
- c) 2, 5, 6, 9;
- d) 3, 8.

*****Following information deals with the questions №194-196;**

The codes of answers:

1. Chemical structure derived from tryptophane;
2. Accelerate growth, germination and development of blossoms;
3. Accelerate the maturation of fruit and drop of foliage;
4. Accelerate growth in height;
5. Inhibits (slows down) the rate of metabolism and development;
6. These are derivates of adenine;
7. Induce cell division in young cells;
8. These are grouped under isoprenoid hormones.

194. Phytohormones influence the growth rate and the development of plants.

Please match cytokinines with the corresponding chemical structure and functions.

- a) 1, 4;
- b) 2, 8;
- c) 7, 6;
- d) 5, 8;
- e) 1, 3.

195. Please match auxines with the corresponding chemical structure and functions.

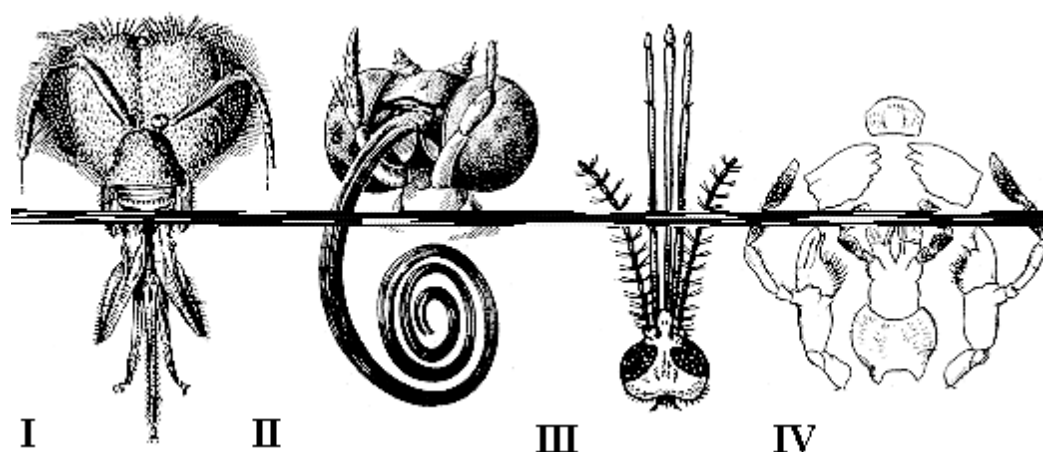
- a) 1, 4;
- b) 2, 8;
- c) 7, 6;
- d) 5, 8;
- e) 1, 3.

196. Please match gibberellines with the corresponding chemical structure and functions.

- a) 1, 4;
- b) 2, 8;
- c) 7, 6;
- d) 5, 8;
- e) 1, 3.

*****Following information deals with the questions №197-200:**

Figures I-IV show different kinds of oral tools common with insects.



197. Which of these tools is specific for butterflies?

- a) I
- b) II
- c) III
- d) IV

198. Which of these tools is specific for honey bee (female worker)?;

- a) I
- b) II
- c) III
- d) IV

199. Which of these tools is specific for cockroach?

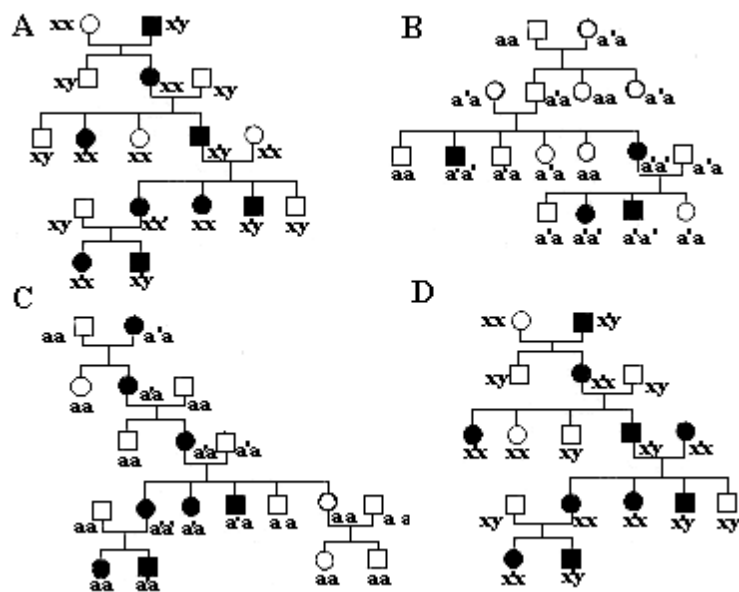
- a) I
- b) II
- c) III
- d) IV

200. Which of these tools is specific for mosquito(female)?

- a) I
- b) II
- c) III
- d) IV

*****Following information deals with the questions №201-204**

Scetch diagrams A-D in which you match the following human hereditary diseases (single gene conditions) with corresponding terms: autosomal, x-linked, dominant and recessive as listed below.



201. Which of the trees correspond to inheritance of the skeletal deformity eg. short fingers (autosomal dominant gene)

- a) A
- b) B
- c) C
- d) D

202. Which of the trees correspond to inheritance of the phenylketonuria and sickle cell anemia (autosomal recessive gene)

- a) A
- b) B
- c) C
- d) D

203. Which of the trees correspond to inheritance of the nystagmism (X-linked dominant gene)

- a) A
- b) B
- c) C
- d) D

204. Which of the trees correspond to inheritance of the red-green colour blindness and haemophilias (X-linked recessive gene)

- a) A
- b) B
- c) C
- d) D.